March 15, 2025

Padma Raghavan, Vice Provost for Research and Innovation
Jonathan Sprinkle, Chair of Computer Science
Vanderbilt University
2201 West End Ave
Nashville, TN 37235

Faisal D'Souza
Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO)
National Science Foundation
2415 Eisenhower Avenue,
Alexandria, VA 22314

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Mr. D'Souza,

Vanderbilt University is grateful for the opportunity to provide input on implementation of President Trump's Executive Order 14179, "Removing Barriers to American Leadership in Artificial Intelligence." We agree with the goals of "sustaining and enhancing America's AI dominance in order to promote human flourishing, economic competitiveness, and national security." To address these challenges, we recommend a three-fold approach:

1. **AI in the Clear**: the AI Action Plan must preserve access to robust AI resources for the researchers who will drive much of the innovation in AI and AI applications.
2. **Digital & AI Literacy**: the AI Action Plan needs to ensure all Americans can understand, develop skills for, and actively participate in shaping the development and use of AI technologies to benefit their communities.
3. **Building on AI-ready Infrastructure**: the AI Action Plan should enable the research, development, and technology transfer needed to deploy innovative AI-powered solutions that address some of our most pressing national priorities, including national security, healthcare outcomes, education and training, and transportation.

Global AI competition is accelerating and the implications for our national security and prosperity could not be greater. AI has the potential to completely upend a wide range of fields; medicine, manufacturing, transportation, biosecurity, cybersecurity and more will change as a result of AI. To ensure the U.S. is both ready for these changes and leading the charge, it is essential to have a whole of government approach that utilizes the public-private partnerships that have been so successful throughout our nation's history. Programs at Vanderbilt University leverage funding from internal funding streams and federal sources such as the National Science Foundation, Dept. of Energy, Dept. of Defense, National Institutes of Health, Dept. of Education, and Dept. of Transportation. We believe it is essential for these and other research agencies to be encouraged to engage in AI research.

AI in the Clear

To avoid being leapfrogged by other countries, **America needs a range of research programs in AI**. Deep Seek and other similar startup companies have demonstrated that even modest infrastructure can produce impressive results in generative AI. Without continued research in this area, America is at risk of being "in the dark" when it comes to anticipating the AI advances of other countries. We recommend OSTP enable programs at all federal research agencies on a range of AI-related subjects, including:

- Foundational AI algorithms and techniques,
- Interdisciplinary AI applications,
- Responsible and trustworthy AI,
- AI ethics and human impacts, and
- The future of AI.

A key element to supporting these research programs is robust AI resources, including access to existing large language models and the computing capability and data necessary for the further advancement of AI. To address the former, Vanderbilt became one of the first universities to provide its students, staff, and faculty access to custom generative AI software; our version is called Amplify. Amplify is Vanderbilt University's internal generative AI platform that leverages the same technology behind ChatGPT and similar AI models. It offers a secure environment, ensuring that data and chat histories remain private and protected within Vanderbilt's internal infrastructure. Amplify also allows the user to easily toggle between existing large language models and benefit from the unique capabilities and strengths of each. With the Amplify program, Vanderbilt University is not just responding to the rapid advancements in AI, it is taking proactive steps to integrate these technologies in a meaningful way. We are also looking for ways to collaborate with other institutions of higher education, to ensure their students, staff, and faculty have access to these tools. **The AI Action Plan should consider mechanisms to scale up these activities.**

The federal government is currently supporting efforts to provide the computational, data, software, model, training and user support resources necessary for the further advancement of AI through the National AI Research Resource (NAIRR) managed by the National Science Foundation. NAIRR is a partnership with 12 other federal agencies and 26 non-governmental partners that makes available government-funded, industry, and other contributed resources in support of the nation's research and education community. NAIRR was originally recommended by a task force that formed as a result of the National AI Initiative Act of 2020 (Public Law No: 116-283), which President Trump signed into law in January 2021. The effort to establish NAIRR had similar motivations as EO 14179 – the need to improve America's AI capabilities to ensure the U.S. remains the leader in this technology. **We urge the AI Action Plan to prioritize support for NAIRR and robustly fund this important program.**

OSTP should ensure the AI Action Plan allows researchers to push the boundaries of foundational machine learning algorithms and architectures. Vanderbilt researchers are leveraging federal funding from agencies such as the National Science Foundation, Dept. of Energy, and Dept. of Defense to advance the state-of-the-art in core AI technologies, such as machine learning algorithms, natural language processing, and computer vision. The public-private partnership is essential for furthering innovation; a few examples of this partnership in action at Vanderbilt include:

- Exploring novel neural network designs, optimization techniques, and learning paradigms to create more powerful, robust, and efficient AI models.
- Advancing the capabilities of language models, enabling better understanding, and generation of human language could lead to significant improvements in applications like intelligent assistants, automated translation, and text analysis.
- Understanding AI's impact of physical systems, such as surgical robots and AI-enabled medical devices, and working to ensure these systems are trustworthy.
- Developing more accurate and versatile visual perception systems, which involves enhancing object recognition, scene understanding, and image/video analysis to support applications in areas like autonomous systems, medical imaging, and surveillance.

Perhaps more important than the contributions of any one field is the importance of the AI Action Plan supporting interdisciplinary collaborations. We have found that interdisciplinary collaborations are a key ingredient in taking the advancements of an individual discipline through to a real-world application. AI has vast potential to address complex real-world challenges, however, without interdisciplinary collaborations, we will struggle to apply these technologies in novel domains. To enable these collaborations at Vanderbilt, we have launched the Vanderbilt Lab for Immersive AI Translation (VALIANT), which serves as a center of gravity for strategic national partnerships and engagement in AI. It will be important for the AI Action Plan to facilitate similar collaborations across federal agencies and the wide range of disciplines they support. At Vanderbilt, we use internal funds to support innovative projects that leverage generative AI to solve problems in areas like healthcare, education, and resource efficiency. By finding similar mechanisms to support this kind of collaboration, the federal government can foster the cross-pollination of ideas and techniques across different domains of AI.

Digital & AI Literacy

American talent is an essential component in winning the global AI competition. **The AI Action Plan must include programs supporting the development of AI skills and knowledge across a wide range of stakeholders**, from current college students to the existing workforce and even K-12 students and teachers. In his EO 13859, signed in February 2019, President Trump said a key principle of the American AI Initiative needed to be training "current and future generations of American workers with the skills to develop and apply AI technologies to prepare them for today's economy and jobs of the future."

Providing upskilling and reskilling opportunities for the current workforce to keep pace with the rapidly evolving field of AI is essential. Vanderbilt's experience in this area is primarily through our relationship with online learning platforms like Coursera to offer a variety of AI-focused courses and professional certificates. We have seen ample interest in these courses, which allows working professionals to develop new AI-related skills and competencies. Based on this experience, we hope an AI Action Plan would identify mechanisms to scale such efforts.

Vanderbilt is also **actively preparing the next generation of AI practitioners and leaders through its academic programs**. Each federal agency has a role to play in educating these students and preparing them with the AI skills that will support the agencies' specific needs. At Vanderbilt, through funding and partnership agreements with DOD – including NSA – and industry, our Institute of National Security is immersing students across campus in AI experiential learning, and our students are discovering new solutions to the nation's most pressing challenges like cybersecurity. The Vanderbilt AI Law Lab is leveraging internal funding to provide law students with hands-on experience in applying AI and machine learning to legal challenges. The university's newly launched Institute for the Advancement of Higher Education (AdvancED) is redefining the faculty and student experience, integrating AI and other emerging technologies across the curriculum to ensure graduates are well-equipped to thrive in an AI-powered workforce. AdvanceED experts are supported by the National Science Foundation, Institute for Education Sciences, National Institutes of Health, and more.

Building on AI-ready Infrastructure

To maximize the potential of AI, the AI Action Plan may need to establish a process by which the federal government determines the sectors, fields, or applications that can most benefit from federal resources. The key goal is to deploy **innovative AI-powered solutions that address some of the most pressing national priorities.** While our researchers are exploring a wide range of application areas, the ones we believe are primed to benefit most and quickly from AI are:

- Strengthening National Security and Defense: Vanderbilt's AI researchers are collaborating with government and military partners to explore how these technologies can be used to enhance situational awareness, decision-making, and mission planning in national security and defense contexts.
- Improving Healthcare Outcomes and Access: Vanderbilt researchers are leveraging AI to enhance medical diagnosis, treatment planning, and patient monitoring. This includes developing AI systems that can analyze medical imaging data and electronic health records to provide clinicians with more accurate and personalized insights.
- Enhancing Education and Training: AI-powered adaptive learning platforms and virtual tutors are being explored at Vanderbilt to personalize the educational experience and improve learning outcomes for students of all ages. Additionally, the university is investigating how AI can be used to create more engaging and effective professional development opportunities for teachers.
- Efficient use of Existing Infrastructure: Vanderbilt is applying AI to the challenge of managing complex transportation systems and infrastructure, such as using computer vision and predictive analytics to improve traffic flow, optimize public transit routes, reduce pedestrian incidents and fatalities, and enhance asset management. Vanderbilt researchers are developing and applying AI to analyze and manage critical infrastructure

(e.g., power, water) risk and improve their reliability and resilience through data-driven and analytical approaches

In addition to supporting the research that results in innovative technologies, **OSTP will need to ensure there are sufficient pathways and support mechanisms to translate that research**. At Vanderbilt, VALIANT has submitted nine patents since it was established seven months ago. This success rate is due in large part to the support structure Vanderbilt has for such technology transfer efforts through our Center for Technology Transfer & Commercialization and The Wond'ry programs. These programs leverage federal support from the National Science Foundation and the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs at most federal research agencies.

Underlying all these application-oriented efforts should be a steadfast commitment to continued basic, fundamental, and curiosity-driven AI research. The federal government is uniquely situated to support basic AI research; even the foundations of our current AI tools were developed at universities with the support of federal funding. By investing in the advancement of core AI technologies and techniques, we ensure a strong foundation for developing innovative and impactful solutions that address real-world problems and create a better future for all.

It will also be important for the AI Action Plan to be centered on ensuring the ethical and responsible development of AI systems. At the core of this point is a commitment – as President Trump said in EO 13859 – "to shaping the global evolution of AI in a manner consistent with our Nation's values, policies, and priorities." One of the principles he laid out in the EO was to "foster public trust and confidence in AI technologies and protect civil liberties, privacy, and American values in their application in order to fully realize the potential of AI technologies for the American people." At Vanderbilt, researchers are working towards the goals of public trust in AI by exploring topics such as algorithm limitations, explainable AI, and the societal impacts of these technologies:

- Algorithms with Limited Training Data: Vanderbilt researchers are actively working to identify and mitigate algorithmic limitations, when AI systems are required to make decisions that are based on limited training sets. This is essential for building trust in AI and ensuring that decisions are in line with human expectations.
- Privacy and Data Rights: Protecting individual privacy and data rights is a paramount concern. Vanderbilt is exploring technical and policy solutions to safeguard sensitive user information throughout the AI system lifecycle.
- Keeping Humans in Charge: While AI has immense potential to augment and empower human capabilities, it is critical that humans retain meaningful oversight and decision-making authority. Vanderbilt is studying ways to preserve human oversight in AI-powered systems while taking advantage of its potential to increase efficiency and save resources.

- Evaluating AI's Impact on America: Vanderbilt recognizes the need to thoroughly assess the broader market impacts of AI deployment. Interdisciplinary research in this area can help policymakers and stakeholders make informed decisions that promote the common good, advance America's competitiveness, and ensure preservation of national security.

In closing, we would point to President Trump's 2019 EO on AI, when he said "The United States must drive technological breakthroughs in AI across the Federal Government, industry, and academia in order to promote scientific discovery, economic competitiveness, and national security." We agree that this principle, as well as the others listed out in that EO, is still relevant today. His EO went on to say we must "Promote sustained investment in AI R&D in collaboration with industry, academia, international partners and allies, and other non-Federal entities to generate technological breakthroughs in AI and related technologies and to rapidly transition those breakthroughs into capabilities that contribute to our economic and national security." We fully agree with the need for sustained investment and continued public-private partnerships to sustain and enhance America's AI dominance.

We appreciate the opportunity to provide input to the AI Action Plan that OSTP was charged with developing in EO 14179. Please let us know what additional information or support we can provide.

Sincerely,

[redacted]

Padma Raghavan, Ph.D.
Vice Provost for Research and Innovation
Jonathan Sprinkle, Ph.D.
Chair of Computer Science

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.